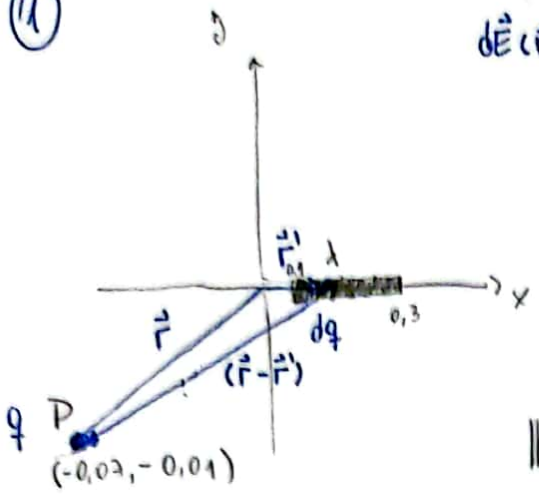
PAUTA - C1 2024 - 1

$\vec{F} = q\vec{E}$ $\qquad 0{,}1 \leq x \leq 0{,}3$

(1)



$$d\vec{E}(\vec{r}) = K\frac{dq}{\|\vec{r}-\vec{r}'\|^3}(\vec{r}-\vec{r}')$$

$$dq = \lambda dx$$

$$\vec{r}' = x\hat{\imath}$$

$$\vec{r} = -0{,}07\hat{\imath} - 0{,}01\hat{\jmath}$$

$$\vec{r}-\vec{r}' = (-0{,}07 - x)\hat{\imath} - 0{,}01\hat{\jmath}$$

$$\|\vec{r}-\vec{r}'\|^3 = \left[(-0{,}07-x)^2 + 0{,}01^2\right]^{3/2}$$

$$\vec{E} = \int_{0{,}1}^{0{,}3} K\frac{\lambda dx(-0{,}07-x)\hat{\imath}}{\left[(-0{,}07-x)^2+0{,}01^2\right]^{3/2}} - \int_{0{,}1}^{0{,}3} K\frac{\lambda dx(0{,}01\hat{\jmath})}{\left[(-0{,}07-x)^2+0{,}01^2\right]^{3/2}}$$

$u = -0{,}07 - x$

$du = -dx$ $\Rightarrow \vec{E} = \displaystyle\int_{-0{,}17}^{-0{,}37} \frac{-K\lambda\, du\, u}{\left[u^2+0{,}01^2\right]^{3/2}}\hat{\imath} - \int_{-0{,}17}^{-0{,}37} \frac{-K\lambda\, du \cdot 0{,}01\hat{\jmath}}{\left[u^2+0{,}01^2\right]^{3/2}}$

$x = 0{,}1 \Rightarrow u = -0{,}17$

$x = 0{,}3 \Rightarrow u = -0{,}37$

$$\vec{E} = K\lambda\left[\frac{1}{\sqrt{u^2+0{,}01^2}}\Bigg|_{-0{,}17}^{-0{,}37}\hat{\imath} + \frac{u}{0{,}01^2\sqrt{u^2+0{,}01^2}}\Bigg|_{-0{,}17}^{-0{,}37}\hat{\jmath}\right]$$

$$\vec{E} = K\lambda\left[(2{,}702 - 5{,}872)\hat{\imath} + (-9996{,}350 - (-9982{,}744))\hat{\jmath}\right]$$

$$\vec{E} = 9\cdot10^9 \cdot 5{,}3\cdot10^{-9}\left[-3{,}17\hat{\imath} - 13{,}606\hat{\jmath}\right]$$

$$\vec{E} = -151{,}209\hat{\imath} - 649{,}006\hat{\jmath}\ [N/C]$$

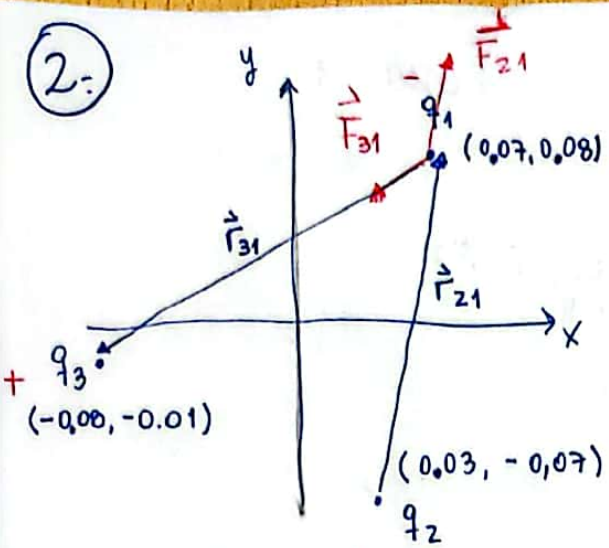
$\Rightarrow \vec{F} = q\vec{E}$

$$\vec{F} = -4{,}5\cdot10^{-6}(-151{,}209\hat{\imath} - 649{,}006\hat{\jmath})$$

$$\vec{F} = 6{,}804\cdot10^{-4}\hat{\imath} + 2{,}921\cdot10^{-3}\hat{\jmath}\ [N]$$

Escaneado con CamScanner

2.-



$$\vec{r}_{21} = (0{,}07 - 0{,}03)\hat{\imath} + (0{,}08 - (-0{,}07))\hat{\jmath}$$
$$= 0{,}04\,\hat{\imath} + 0{,}15\,\hat{\jmath}$$

$$\vec{r}_{31} = (-0{,}08 - 0{,}07)\hat{\imath} + (-0{,}01 - 0{,}08)\hat{\jmath}$$
$$= -0{,}15\,\hat{\imath} - 0{,}09\,\hat{\jmath}$$

$$\|\vec{r}_{21}\|^3 = [0{,}04^2 + 0{,}15^2]^{3/2}$$
$$= 3{,}741 \cdot 10^{-3}$$

$$\|\vec{r}_{31}\|^3 = [(-0{,}15)^2 + (-0{,}09)^2]^{3/2}$$
$$= 5{,}353 \cdot 10^{-3}$$

$$\vec{F}_1 = \vec{F}_{31} + \vec{F}_{21}$$
$$= K \frac{|q_1||q_3|}{\|\vec{r}_{31}\|^3}\vec{r}_{31} + K \frac{|q_1||q_2|}{\|\vec{r}_{21}\|^2}\vec{r}_{21}$$

$$\vec{F}_1 = \left[9 \cdot 10^9 \frac{(3 \cdot 10^{-6})(6 \cdot 10^{-6})}{5{,}353 \cdot 10^{-3}}\right](-0{,}15\,\hat{\imath} - 0{,}09\,\hat{\jmath})$$
$$+ \left[9 \cdot 10^9 \frac{(3 \cdot 10^{-6})(4 \cdot 10^{-6})}{3{,}741 \cdot 10^{-3}}\right](0{,}04\,\hat{\imath} + 0{,}15\,\hat{\jmath})$$

$$= -4{,}540\,\hat{\imath} - 2{,}724\,\hat{\jmath} + 1{,}155\,\hat{\imath} + 4{,}330\,\hat{\jmath}$$

$$\boxed{\vec{F}_1 = -3{,}385\,\hat{\imath} + 1{,}606\,\hat{\jmath} \ [N].}$$

Escaneado con CamScanner